

Web Technologies 1 (Frontend)

Week 2: HTML Advanced

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1 Overview

Week 2 goes beyond the basic HTML skeleton: tables, forms/inputs/buttons, lists/links/images, semantic tags, `<div>` and `<span>`, and finally how to publish a site with GitHub Pages.

2 1. Tables

Tables organize related information into rows and columns.

```
<table>
  <thead>
    <tr><th>Subject</th><th>Day</th></tr>
  </thead>
  <tbody>
    <tr><td>Web Tech 1</td><td>Monday</td></tr>
    <tr><td>Algorithms</td><td>Tuesday</td></tr>
  </tbody>
  <tfoot>
    <tr><td colspan="2">Total: 2 classes</td></tr>
  </tfoot>
</table>
```

- `<tr>` — table row
- `<th>` — table header cell
- `<td>` — table data cell
- `colspan` lets a cell span multiple columns.

3 2. Forms, Inputs and Buttons

The `<form>` element collects user input.

```

<form>
  <label>Name</label>
  <input type="text">

  <label>Email</label>
  <input type="email">

  <button>Send</button>
</form>

```

- An `<input>` element's appearance/behavior depends on its `type` attribute.
- Common input types: `text`, `email`, `password`, `number`, `date`, `color`, `checkbox`, `radio`, `file`, `submit`.
- `<label>` defines a caption for a form control (improves accessibility and click targets).

4 3. Lists, Links and Images

```

<ol> <!-- ordered list -->
  <li>Football</li>
  <li>Chess</li>
</ol>

<ul> <!-- unordered list -->
  <li>
    <a href="https://example.com">Visit my website</a>
  </li>
</ul>



```

- `<ol>` — ordered (numbered) list; `<ul>` — unordered (bulleted) list; `<li>` — list item.
- `<a href="...">` creates a hyperlink.
- `` embeds an image; `alt` provides fallback/accessible text.

5 4. Semantic Tags

- **Non-semantic elements** (`<div>`, `<span>`) tell nothing about their content.
- **Semantic elements** (`<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<aside>`, `<footer>`) clearly describe their meaning to both the browser and the developer.

- `<header>` — top of the page: logo, title.
- `<nav>` — navigation links.
- `<main>` — the unique content of the page.
- `<section>` — a thematic block with a heading.
- `<article>` — a self-contained piece: a post, a card.
- `<aside>` — side content: menu, related links.
- `<footer>` — contacts, copyright.

6 5. `<div>` and `<span>`

- `<div>` — a block container: takes the full width available, always starts on a new line, groups elements so CSS can style them together.
- `<span>` — an inline container: takes only the width it needs, stays inside the line of text, used to style a few words inside a paragraph.

```
<div class="card"> ... </div>

<p>today is <span class="hot">98</span></p>
```

7 6. Publishing with GitHub Pages

1. Register on GitHub (<https://github.com/>).
2. Create a new repository.
3. Upload website files — the project's main HTML file must be named `index.html`.
4. Enable GitHub Pages in the repository Settings.
5. Access the published site at the provided URL: `your-username.github.io/your-repo-name`.

8 Summary

Week 2 extends HTML with structured data (tables), user input (forms/inputs/buttons), navigation content (lists, links, images), meaningful page structure (semantic tags), generic grouping containers (`div/span`), and the practical step of publishing a static site for free via GitHub Pages.